



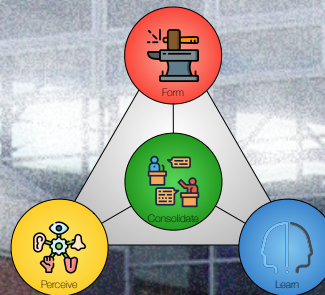
AALBORG UNIVERSITY
DENMARK

6 HORIZON

ESSENCE – CHAPTER 9

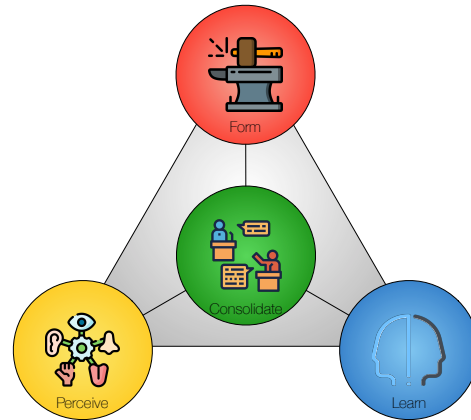
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SWI 2026



Agenda

- Exercise 5 Follow-up.
- Chapter 9: Horizon.
 - Pathways—Dewyan Pragmatism.
 - Horizons.
 - Technology Push.
 - Market Pull.
 - Innovation Types.
 - Horizon Hypotheses.
- Next time:
 - Exercise 6: Horizon.
 - Lecture 7: Strategy & Outer Environment.

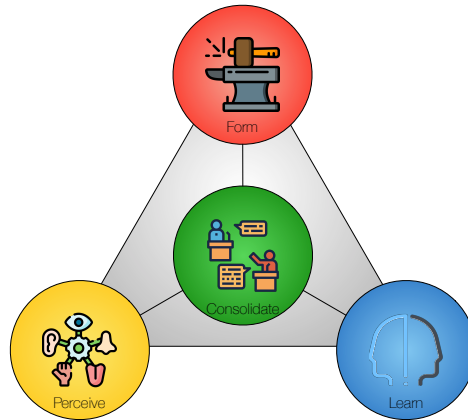


Follow-up: Exercise 5

Exercise 5 (Solution Scenarios)

Use the results from **Exercise 4** in this exercise.

- Develop **axes** for the Solution Scenarios as described in Section 8.3 (Tables 8.2 and 8.3).
- Develop **Solution Scenarios** as illustrated in Figure 8.1.
- Explain the **Solution Scenarios** you find in each quadrant.
- Choose one scenario to work with from now on. You can choose one quadrant or combine several if relevant.
- Discuss short-term **strengths** and **weaknesses** of your chosen scenario—how might this scenario create value and growth in the short run?
- Discuss longer-term **opportunities** and **threats** related to your chosen scenario—how might the future look if you use this scenario?

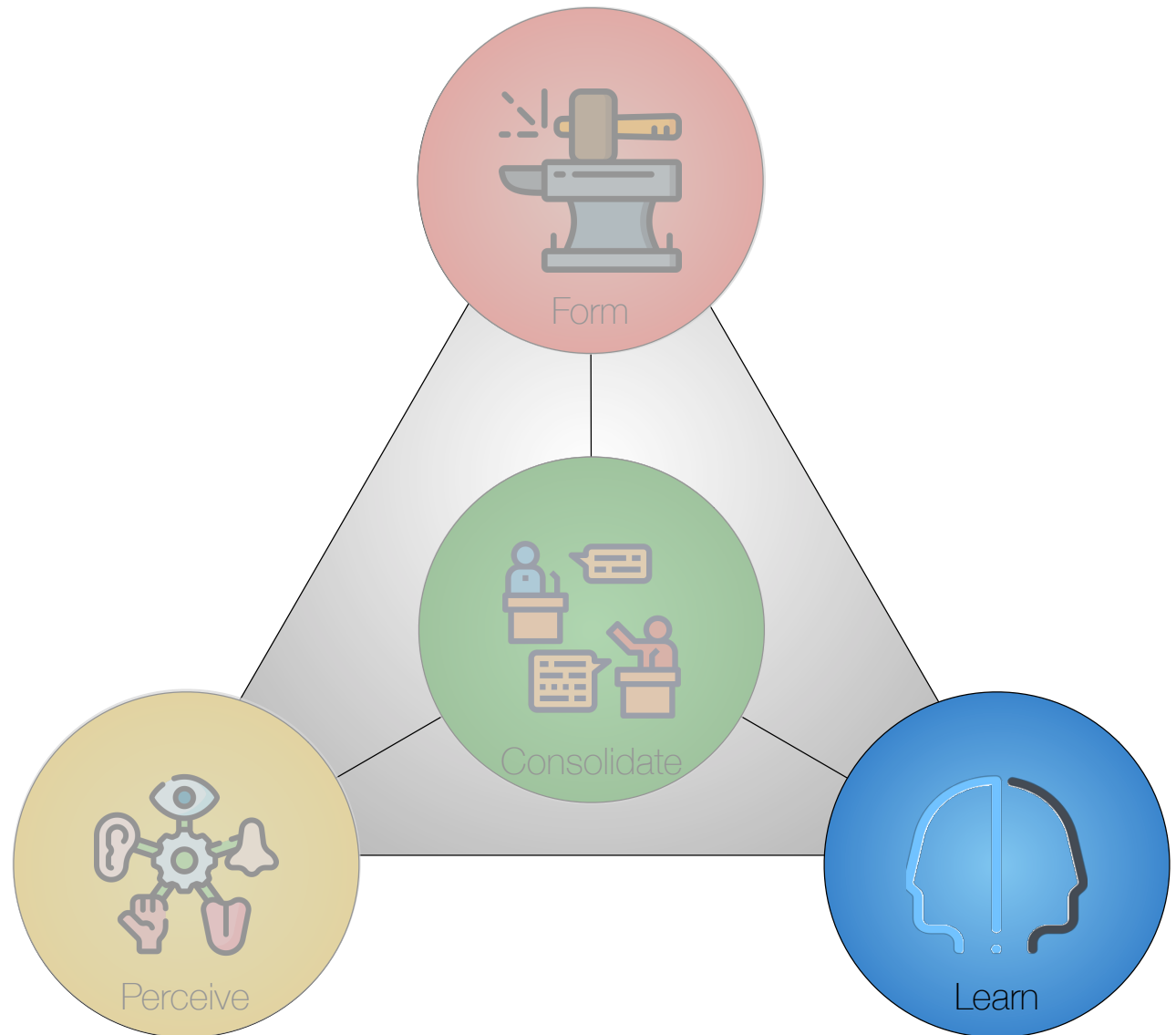


Horizon

We choose to go to the Moon [...] because that goal will serve to organize and measure the best of our energies and skills, because that challenge is one that we are willing to accept, one we are unwilling to postpone, and one we intend to win

Four Core Activities

- *Perceive* – understanding the problem and its context.
- *Form* – designing the constructive parts of a solution.
- *Consolidate* – integrating the design contributions into a clear whole that addresses the situation.
- *Learn* – appraising the design.



Three Levels of Abstraction



Rationale – the logical basis that makes actions meaningful and helps reason about strategy and tactics for solving the overall problem.



Strategy – the master plan for solving the overall problem, including the scope of the problem, the key components for building the solution, and qualifications regarding limitations and constraints that may affect the utility or acceptability of a solution.



Tactics – actions planned in accordance with the strategy to achieve specific ends.

Transformational Growth

While infrastructure is of course important, it must be part of a bigger **vision**.

The point is not to simply dig ditches but to **steer** those investments towards transformational growth.

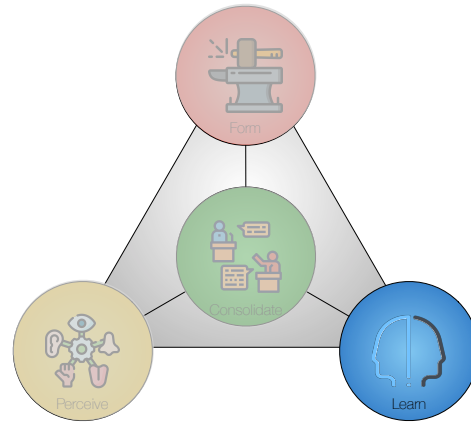


Mariana Mazzucato

A Question of Mindset

*Essence is aimed at design teams and projects; however, although the scale of such activities is much smaller than that of the Apollo Mission, we propose that a similar **mindset** is necessary today.*

*We must view each project not only as a way to address the immediate problem but also as an **opportunity** to develop **strengths** that will help tackle **future challenges**.*



Pathways

Deweyan Pragmatism

Where are we?

Are we getting closer?

*An idea of an **end** to be reached, an **end-in-view**, is logically indispensable in discrimination of existential material as the evidential and testing **facts** of the case.*

*Without it, there is no **guide for observation**; without it, one can have no conception of what one should look for or even is looking for.*

*One "**fact**" would be just as good as another – that is, good for nothing in control of inquiry and formation and in settlement of a problem.*

Appreciative judgment

There is no inquiry that does not involve judgments of practice.

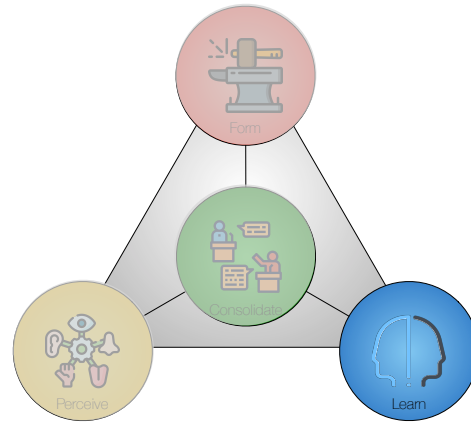
- We look for **evidence** to decide if our goals have been met.
- **Goals** represent our **understanding** of what must change to resolve the problem.
- The evidence must therefore help decide – on grounds of **existential** and **observable** changes – if a determinate situation has been reached.

End-in-views in Essence

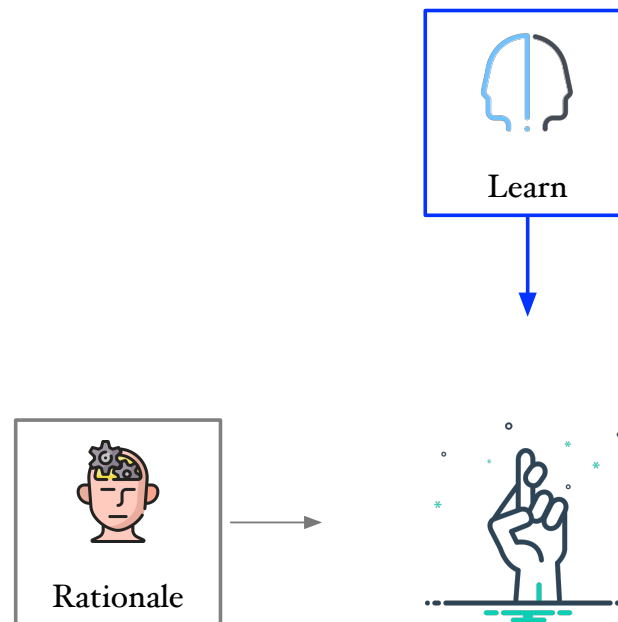
In Essence, we have two types of end-in-view:

- *Mission end-in-view*, which pertains to the **current** project.
- *Horizon end-in-view*, which focuses on preparing for a **future** beyond the current project.

Both types can be **distant** and **subject to change**, as they may be elusive and outside our scope of knowledge.



Horizons

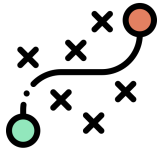


Horizons in Essence

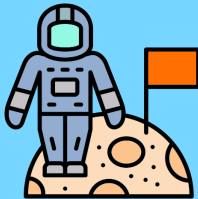
In Essence, we have three horizons:

- The first horizon (**H1**) represents **business as usual**—the way things are done now.
- The second horizon (**H2**) represents the **shift** and **clash** of different ideas about the future.
- Horizon Three (**H3**) is long-term and represents an **emerging** future pattern, including **technologies**, **application areas**, **norms**, **values**, and **economics**.

Learn Terminology



Mission



The mission presents the project's tactical objective: to solve the problem.

Potential



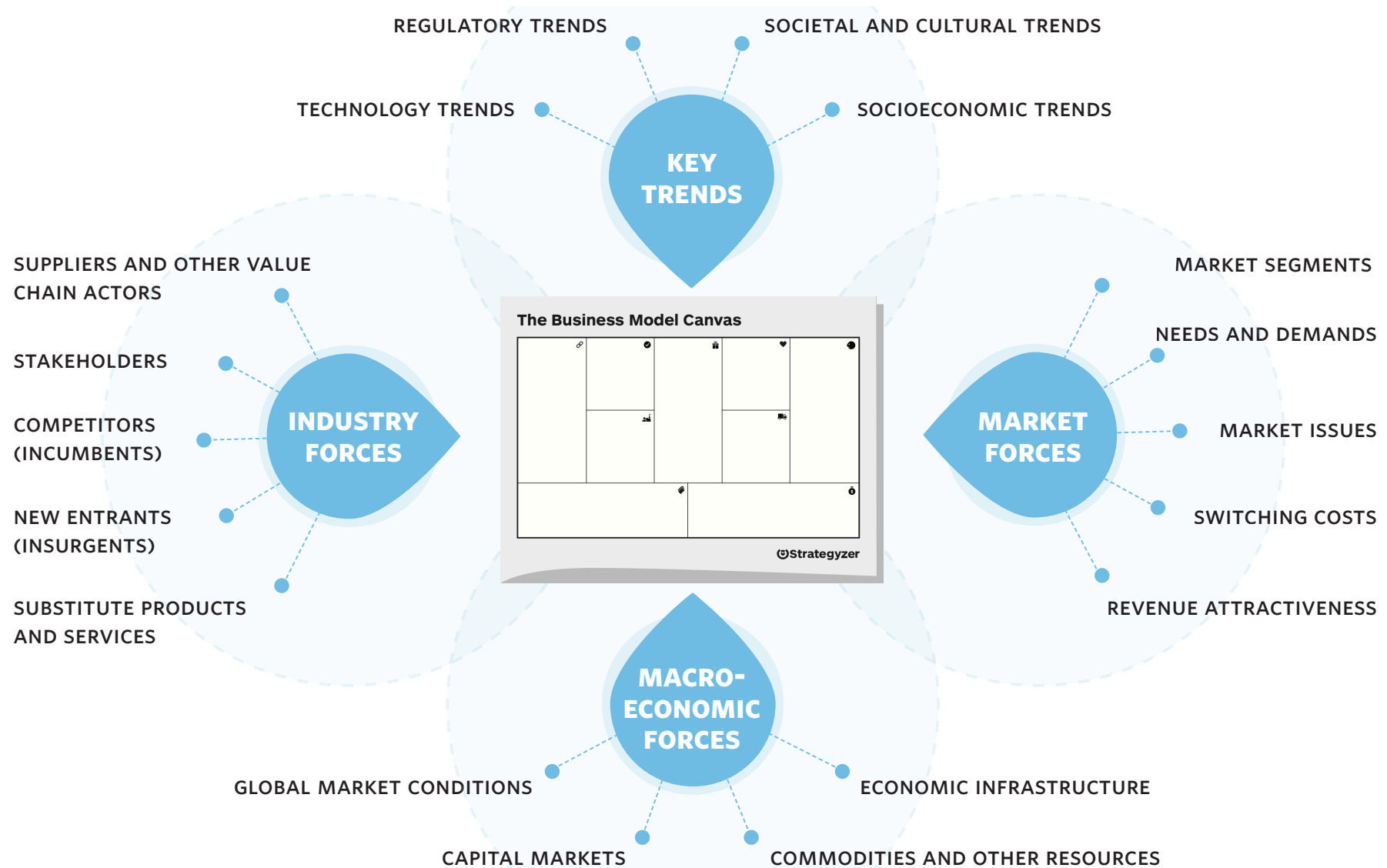
The potential defines the project's strategic objective: to create strengths and opportunities.

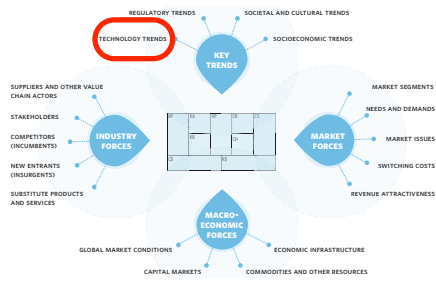
Horizon



Where might the outcomes of this project lead us? How do they align with the future we aim to be a part of?

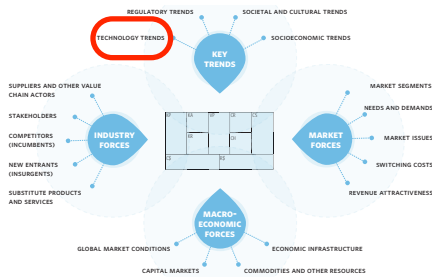
Business Model Environment





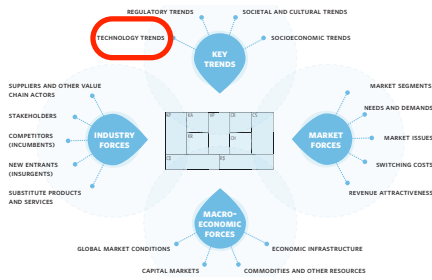
Technology Push

- Technology and science as **drivers of change**.
- Should we base **future leverage** on advancements in **technology and science**?



Technology Push 2

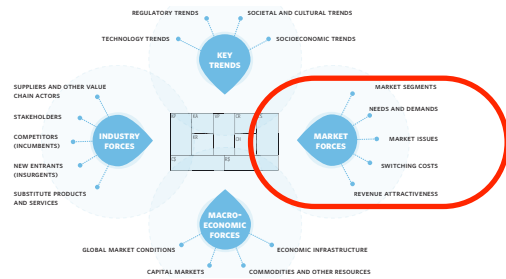
- We have witnessed **significant advancements** in artificial intelligence, quantum computing, virtual reality, augmented reality, digital platforms, and more, and this **trend** will likely persist indefinitely.
- Will these developments affect any of the types of **leverage** we currently have? Will **new leverage points** develop, or will **existing ones** be threatened? Will **we** be able to handle new challenges better than **others** will?



Technology Push 3

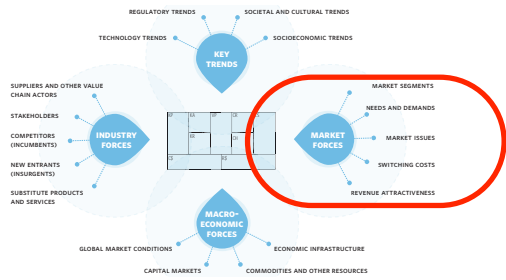
For example:

- Will **artificial intelligence** be used to handle complex **coding** tasks, suggest more innovative code, or assist in **refactoring** intricate systems?
- Will the increased use of **microservice**-based architectures provide significant advantages?
- Can we create solutions with improved **accessibility**—new ways to interact without physical controls?
- Can we utilize **brain-computer interfaces (BCI)**—to develop new leverage points?



Market Pull

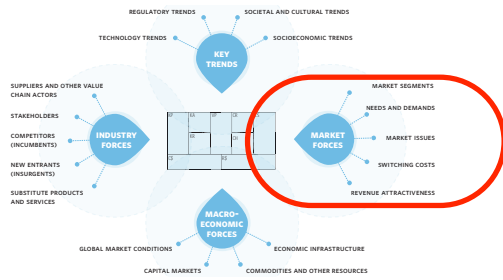
- Market features, users, economic developments, and more as **drivers of change**.
- Should we base **future leverage** on significant trends and changes in forces in the **market, industry, and economy**?



Market Pull 2

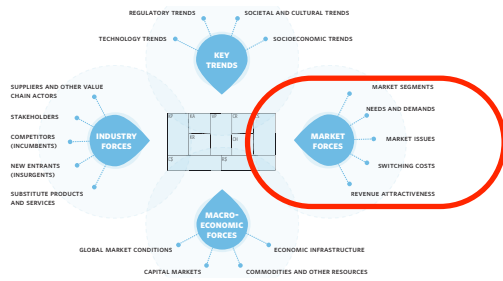
There have been **significant developments** in

- **Business practices**—industry organization, production locations and methods, logistics.
- **Globalization**—international and national regulations, income distribution, trade, tariffs
- A **growing middle class** with greater purchasing power. More focus on crime, security, and surveillance.
- New forms of **warfare** and **new strategies** for conducting them.
- Dependencies on **critical infrastructure** and ways to protect and ensure **resilience**.



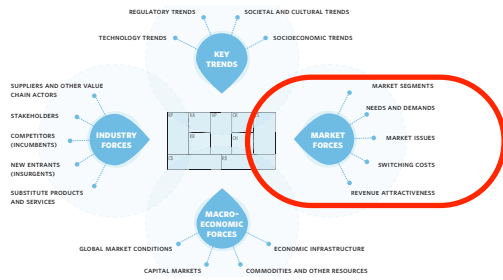
Market Pull 3

- Will these developments create **new opportunities** or **diminish the needs** we have catered to thus far?
- Will **new markets** emerge, or will **existing ones** be at risk? Will we be able to **target** these markets more effectively than others?



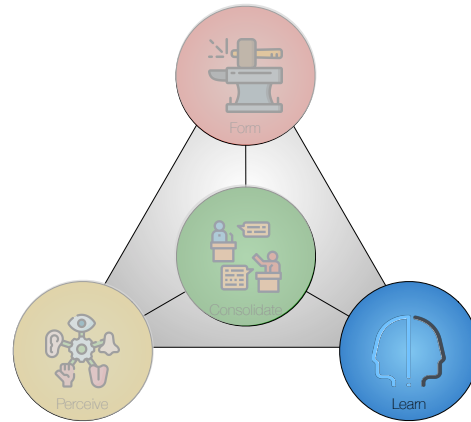
Innovation Types

- Product, process, position, and paradigm are four different variants of innovation.
- Will we enhance our leverage by changing the kind of innovation?



Innovation Types 2

- Will changes in **technology** or **demands** invite changes in what we offer?
- If we provide products, will we **continue to do so**, or **will we shift** to another type of offer?
- What **contributions** will allow us to **excel** in the future? Will they be **products**, **services**, or **processes**?



Horizon Hypotheses



Purpose of Hypotheses

- We use **hypotheses** to ensure our design serves its **intended purpose** and meets our expected **standards** and **end-in-view**.
- A hypothesis is a **proposition** that gives us a basis for **reasoning** about something we need to **investigate** further.
- It is a starting point for **examining** whether the contribution **conforms** to the **expected** solution.
- Hypotheses are **verifiable** or **falsifiable** and predict specific outcomes.

Hypotheses and Criteria

- Hypotheses are the basis for reviewing.
- Their formulation during the design process helps clarify the project's goals.
- To explain these goals, we must specify what to look for.
- For that reason, we must develop criteria.

Rationale Hypotheses

- **Problem:** We hypothesize that the problem is well understood and defines the difficulty we must overcome.
- **Leverage:** We hypothesize that we employ technologies, components, information, or human resources that make our solution stand out.

What criteria could check such hypotheses?

Rationale Hypotheses 2

- **Prospect:** We hypothesize that the prospect solves the problem. This is inspired by what Ries calls a **leap of faith**. If it fails, the design makes no sense.
- **Warrant:** We hypothesize that solving the problem is valued by stakeholders, customers, and users. This is inspired by what Ries calls the **growth hypothesis**: The solution will likely spread.
- **Backing:** We hypothesize that the prospect is an attractive and valuable solution. This is inspired by what Ries calls the **value hypothesis**: The proposed solution is attractive and worth the costs.

What criteria could check such hypotheses?

Horizon Hypotheses

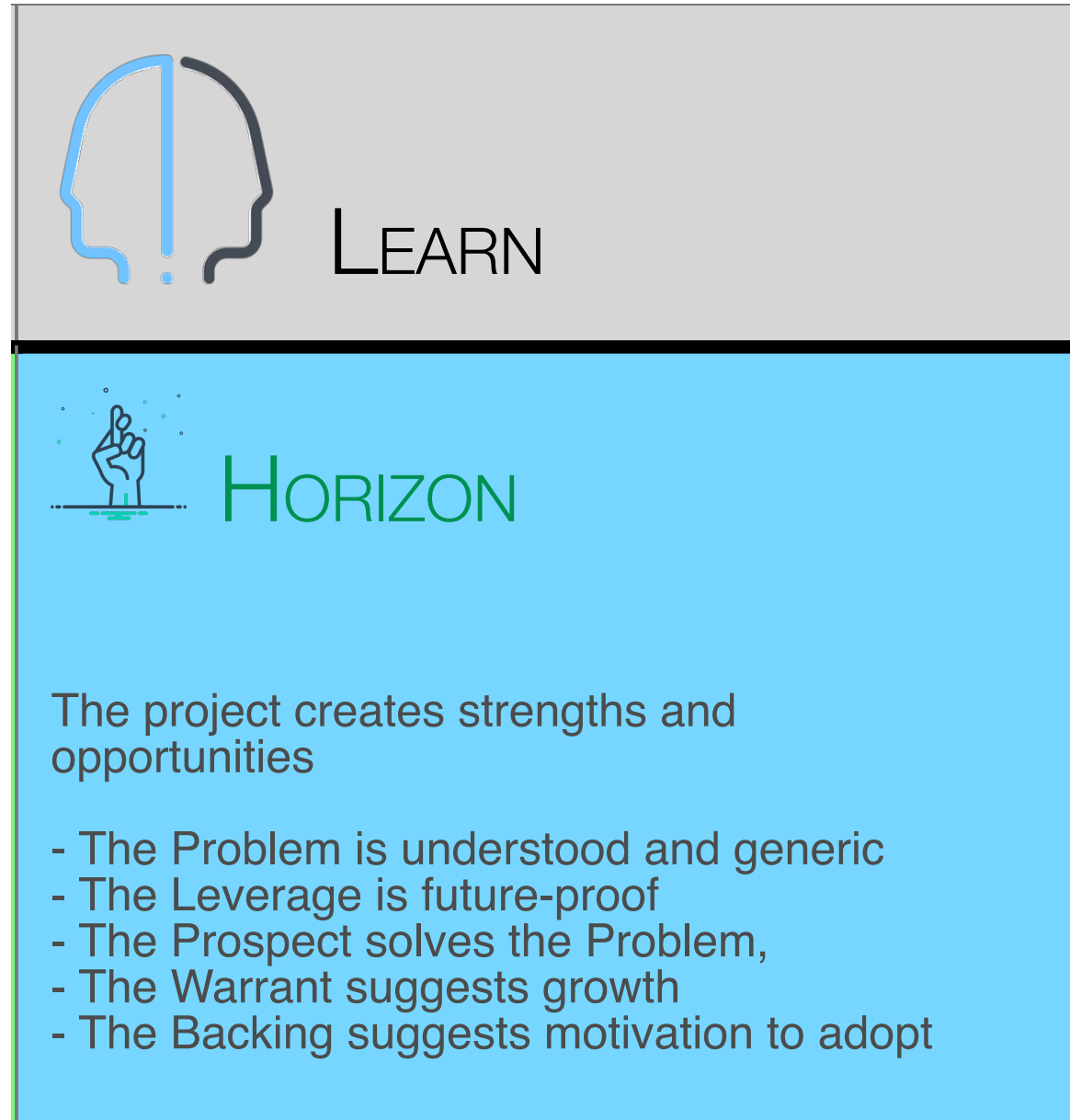
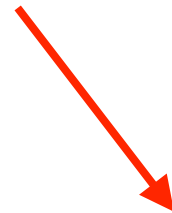
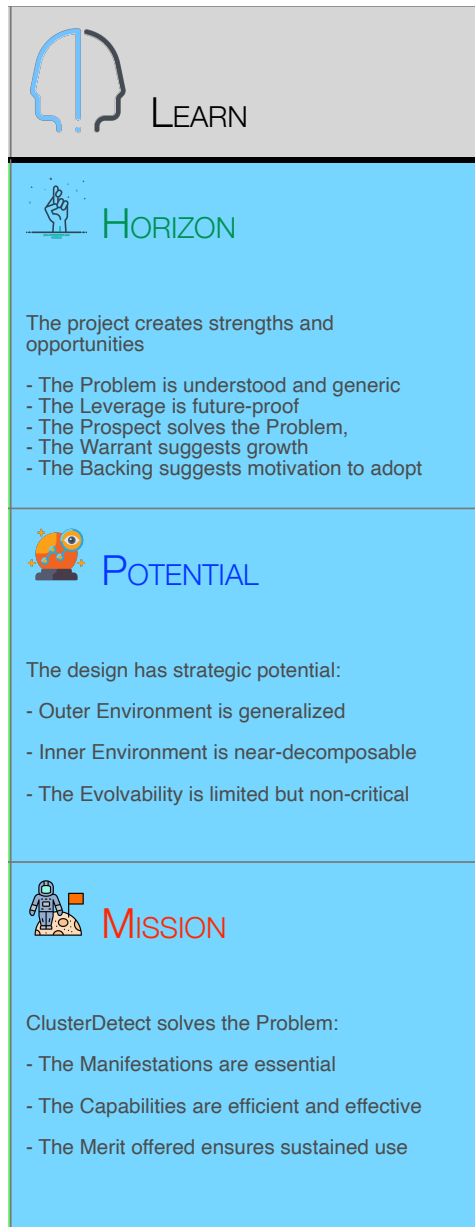
Mission end-in-view

Horizon end-in-view

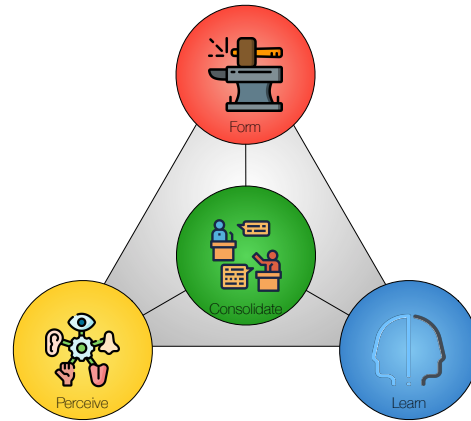
TIME	THE PRESENT	THE FUTURE
PROBLEM	The problem covers the issues faced by the stakeholders.	This problem will be significant in the future.
LEVERAGE	The leverage is vital for this project.	The leverage developed will help solve future challenges.
PROSPECT	The prospect solves the problem well.	This prospect is relevant to a category of future challenges.
WARRANT	The problem is worth solving.	This solution will help address future problems worth solving.
BACKING	The solution is effective and competitive.	This project will facilitate effective and competitive future solutions.

Table 9.1: Horizon hypotheses

Illustration: ClusterDetect



Combining horizon and mission end-in-view



Next Time

Next Exercise and Lecture

- Exercise 6:
 - Horizon.
- Lecture 7:
 - Chapters 10 & 11 in *Essence: Strategy and Outer Environment*.

Exercise 6 (Horizon)

Use the results from **Exercise 5** in this exercise. This exercise is about the **project's horizon**.

- Will your horizon be affected by the **technology push**? What and how?
- Will your horizon be affected by **market pull**? What and how?
- Will changes in **innovation type** characterize your horizon? How and why?
- Formulate **horizon hypotheses** as outlined in Table 9.1. Make sure you consider the future as well as the present.
- Outline some **criteria** you might use to verify or falsify your **hypotheses**.